Department of Computer Science & Engineering



NEXUSEDU AI

Full-Stack Student & Campus Intelligence Platform with Unified Dev Matrix & AI Placement Diagnostics

MINI PROJECT | SESSION 2026-2027

PRESENTED BY: TEAM NO. 12

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NAVIGATION ROADMAP

PROJECT PRESENTATION OUTLINE

01 Introduction
Domain background, real-world problem statement & project motivation

02 Objectives & Scope
Concrete action goals, core deliverables & technical scope limits

03 Existing vs Proposed System
Critical comparison with traditional LMS and literature gaps

04 Technology Stack
Frontend, Flask 3.1 Python backend, SQLite3 & Gemini AI engine

05 Resource Requirements
Hardware specifications & software tools version matrix

06 System Design & Architecture
3-Tier system topology, data flow (DFD) & relational schema

07 Methodology
14-Week Agile iterative lifecycle & sprint milestones

08 Modules & Features
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09 Implementation & Results
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Unit, functional, security, session bypass & cross-browser tests

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Academic papers, official documentation & RFC citations (APA Style)

01 INTRODUCTION

INTRODUCTION: DOMAIN, PROBLEM & MOTIVATION

1. Domain & Background

Full-Stack Web Engineering & Developer Analytics

- Focuses on building modern, responsive, high-performance web applications integrated with relational databases and machine learning/AI APIs.
- In today's software recruitment ecosystem, campus placements demand multi-dimensional technical excellence: strong DSA problem-solving and clean Git version control.
- Educational institutions face significant challenges tracking these decentralized activities without a unified platform.

2. Real-World Problem Statement

Fragmented Developer Profiles & Manual Academic Overheads

- Students practice coding across disconnected platforms (LeetCode, GitHub, College LMS), leading to fragmented progress and lack of holistic visibility.
- Faculty members endure high grading burdens, manually reviewing code archives and tracking assignments on spreadsheets without standardized rubrics.
- Lack of continuous feedback leads to student burnout and late-stage realization of industry skill gaps.

3. Motivation & Proposed Impact

Bridging the Academia-Industry Skill Gap

- NexusEdu AI bridges this chasm by delivering a single-pane-of-glass dashboard uniting college academics with live external developer metrics.
- Combines real-time algorithmic telemetry with a gamified ByteCoin virtual economy to sustain daily coding streaks.
- Equips faculty with AI-assisted coursework drafting and automated rubric evaluation, modernizing college governance.

02 OBJECTIVES & SCOPE

PROJECT OBJECTIVES & FUNCTIONAL SCOPE

CORE PROJECT OBJECTIVES

• To design and deploy a unified 90-day activity matrix synchronizing LeetCode DSA solves and GitHub repository commits day-by-day.
• To develop a multimodal AI Placement Readiness diagnostic engine calculating dynamic composite scores (0-100%) and personalized 7-day roadmaps.
• To construct a gamified virtual economy with XP level curves, ByteCoins currency, daily streaks, cosmetic customizers, and celebratory chimes.
• To build a dedicated Faculty Portal enabling targeted coursework publishing, AI rubric auto-grading, and campus bulletin broadcasting.
• To implement strict session-based authentication preventing bypasses via cryptographic cookies and role guards (@faculty_required).

SYSTEM BOUNDARIES & SCOPE

Included in Current Release (v2.4)

- Full-Stack Web Architecture on Flask 3.1 and SQLite3 database.
- 90-Day multi-month calendar with viewport floating tooltips.
- Interactive AI Code Arena (Roast, Turbo Optimize, Edge Cases).
- FAANG Mock Interview Challenge Simulator with XP rewards.
- Coursework submission queue with file uploads and faculty grading.
- Campus circular broadcasting with priority urgency alerts.

Not Included / Future Scope Limitations

- Native Android / iOS mobile applications (currently responsive web).
- Automated plagiarism AST parsing against external global repos.
- Multi-university federated single sign-on clustering.

03 COMPARATIVE ANALYSIS

COMPARATIVE ANALYSIS: EXISTING vs PROPOSED SYSTEM

System Attribute	Existing Traditional Approach	NexusEdu AI Proposed System
Developer Tracking	Scattered profiles across LeetCode, GitHub, HackerRank; no holistic correlation.	Unified 90-Day dual synchronized activity heatmap with real-time day inspector.
Placement Analytics	Subjective assessments, manual resume reviews, unquantified placement readiness.	Multimodal AI Readiness Index (0-100%) synthesizing DSA, Git commits & coursework.
Student Engagement	Static assignment submission deadlines, low motivation, isolated coding practice.	Gamified ByteCoin economy, progressive leveling (1-50), daily streaks & shop customizers.
Faculty Workload	Manual grading queues, paper/spreadsheet evaluations, zero rubric scaffolding.	AI-assisted assignment drafting, automated rubric suggestions, 1-click evaluation.
Security & Access	Unsecured cookies, client-side auth bypasses, mixed role permissions.	Strict cryptographic HTTPOnly session cookies, role guards (@faculty_required, @student_required).

Literature Gap Summary:
While traditional LMS solutions (e.g. Moodle, Canvas) manage coursework files and coding portals (LeetCode, HackerRank) track competitive problem solving, no existing commercial or open-source tool unifies academic submissions with verified external Git commits and algorithmic growth under a single gamified placement readiness intelligence model.

04 TECHNOLOGY STACK

FULL-STACK TECHNOLOGY ARCHITECTURE

FRONTEND TIER

- **HTML5 & Modern CSS3**
  Semantic layouts with custom CSS properties
- **Tailwind CSS Engine**
  Glassmorphism & Cyberpunk dark design tokens
- **Vanilla JavaScript ES6+**
  Lightweight DOM manipulation without heavy frameworks
- **Chart.js & Canvas Visuals**
  Doughnut charts, progress bars & confetti particles
- **Web Audio API Chimes**
  Synthetic procedural audio feedback on level-ups

BACKEND TIER

- **Python 3.13 Runtime**
  Fast, modern execution environment
- **Flask 3.1.0 Framework**
  Lightweight, decoupled WSGI micro-framework
- **RESTful JSON APIs**
  Clean endpoint routing across 20+ specialized routes
- **Werkzeug Security**
  Cryptographic password hashing (Scrypt & SHA256)
- **Session Access Guards**
  Decorator-driven RBAC (@login_required, @faculty_required)

DATABASE & ORM

- **SQLite3 Relational DB**
  Serverless, zero-configuration ACID storage
- **Normalized Schema**
  6 relational tables with foreign key cascades
- **Row-Factory Mapping**
  Fast dictionary-style record serialization
- **Transactional Safety**
  Atomic rollback on submission & store purchases
- **Automated Migrations**
  Schema initialization script ensuring zero preloaded dummy data

AI & CLOUD PLATFORM

- **Google Gemini 2.0 Flash**
  Multimodal intelligence API for real-time analysis
- **google-genai SDK**
  Official modern SDK with fast model fallbacks
- **Local Reasoning Engine**
  Pattern-aware AST analysis for code review & Big-O
- **Firebase Client Auth**
  Google OAuth 2.0 popup integration & ID tokens
- **Local Session Bridge**
  Seamless authentication syncing between Firebase & Flask

05 RESOURCE REQUIREMENTS

HARDWARE & SOFTWARE SPECIFICATIONS

1. Hardware Specifications

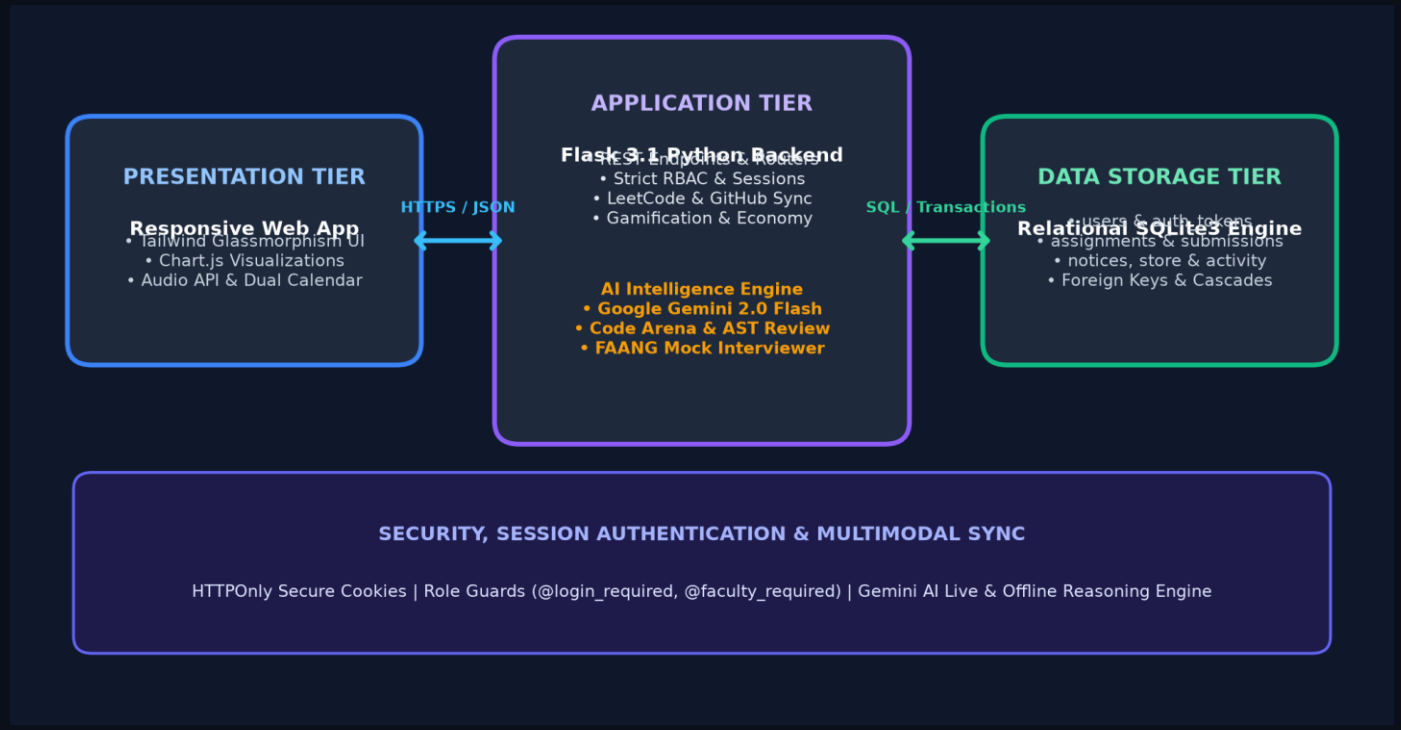
Component	Minimum Requirement	Recommended Spec
Processor	Intel Core i3 / AMD Ryzen 3 (2.0 GHz)	Intel Core i5/i7 (11th Gen+) or Ryzen 5+
Memory (RAM)	4 GB DDR4	8 GB / 16 GB DDR4/DDR5
Disk Storage	256 GB HDD / SSD (500 MB free)	512 GB NVMe SSD for fast compilation
Network Interface	Standard Wi-Fi / LAN (2 Mbps)	High-Speed Broadband (10+ Mbps)

2. Software Tools & Versions

Software / Tool	Version Tested	Role in Project
Operating System	Windows 11 (24H2) / Ubuntu 24.04	Development Host OS
Python Runtime	Python 3.13.0 (64-bit)	Core Backend Runtime
Flask Framework	Flask 3.1.0 + Werkzeug 3.1	Web Server & REST APIs
Database Engine	SQLite3 (Relational ACID)	Structured Data Persistence
AI / GenAI SDK	google-genai 1.0+ (Gemini)	Multimodal Placement AI

06 SYSTEM ARCHITECTURE

SYSTEM DESIGN & 3-TIER ARCHITECTURE

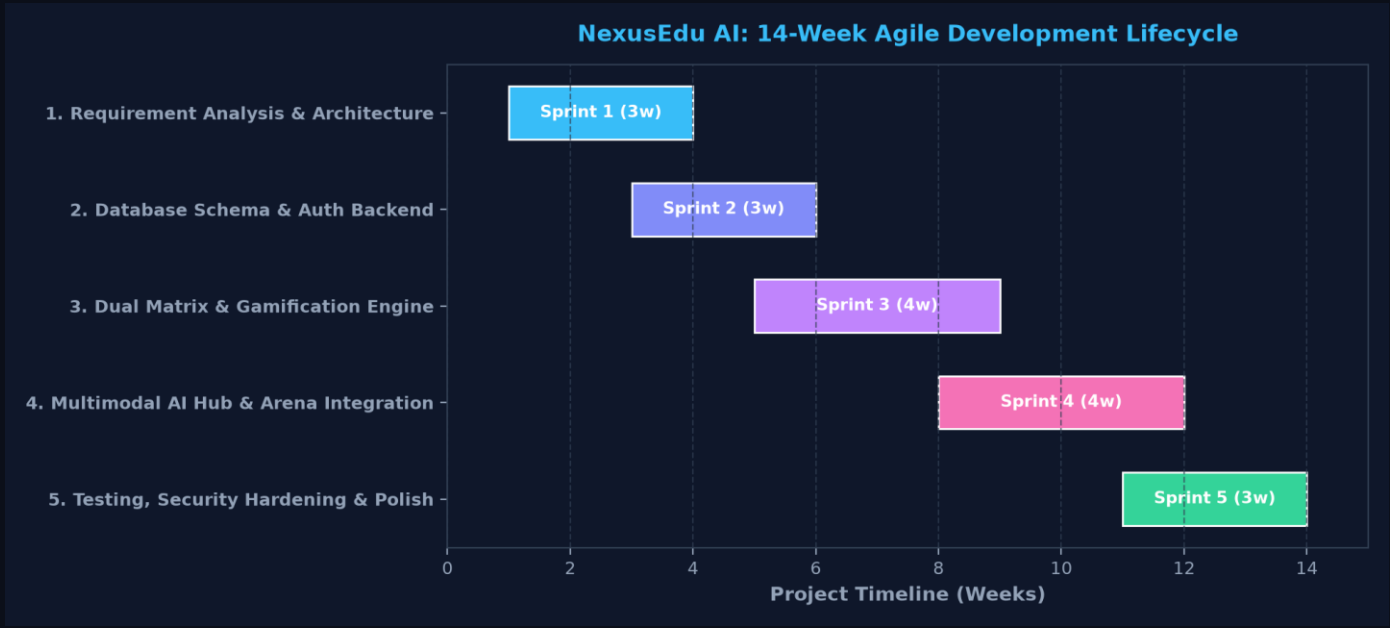


KEY ARCHITECTURAL HIGHLIGHTS

- **Presentation Tier Decoupling:**
  Client interacts solely through asynchronous JSON fetch calls. Zero full-page reloads for gamification, shop, and calendar updates.
- **Layered Role-Based Access:**
  Route protection enforces separation of privileges: Students cannot view the grading queue; Faculty cannot bypass administrative dashboards.
- **Dual-Engine AI Redundancy:**
  Integrates Google Gemini 2.0 Flash for cloud reasoning with an AST-driven local heuristic engine when offline or network-constrained.
- **Normalized Relational Persistence:**
  ACID transactional guarantees prevent corrupted balances when purchasing items in the ByteCoins cosmetic shop.

07 METHODOLOGY

AGILE METHODOLOGY & SPRINT MILESTONES



AGILE ITERATIVE MODEL

- **Sprint 1: Inception & SRS:** Gathered user requirements from 2nd-year CS students and faculty members. Finalized data dictionary and UI wireframes.
- **Sprint 2: Auth & Database Core:** Implemented SQLite relational schema and strict session-based authentication with role-based routing.
- **Sprint 3: Dual Matrix Heatmap:** Engineered 90-day multi-month calendar syncing LeetCode problems and GitHub commits into unified date tiles.
- **Sprint 4: Multimodal AI & Gamification:** Integrated Google Gemini 2.0 Flash, AI Code Arena, FAANG Mock Interviewer, and the ByteCoins virtual shop.
- **Sprint 5: Validation & Deployment:** Conducted automated test suites, fixed HTML attribute overflows, and packaged pitch-ready deliverables.

08 MODULES & FEATURES

CORE SYSTEM MODULES & DELIVERABLES

MODULE 01: User Authentication & RBAC
- Dual Firebase Google OAuth & local Scrypt fallback.
- Cryptographic session cookies with role guards.
- Demo 1-click bypasses permanently blocked.

MODULE 02: Unified 90-Day Activity Matrix
- Synchronizes LeetCode solves (⚡) and Git commits (🔥).
- 7-column calendar with month navigator tabs.
- Global floating viewport tooltips (never clipped).

MODULE 03: AI Diagnostic Engine & Arena
- Placement readiness score (0-100%) across DSA, Git & OS.
- AI Code Arena: Roast, Turbo Optimize & Edge Cases.
- FAANG Mock Interview Challenge Simulator.

MODULE 04: Gamification & ByteCoins Economy
- Level 1 to 50 progression curve with verified XP.
- Daily streaks with claim cooldowns & ByteCoins payouts.
- Cosmetic shop: Cyberpunk avatars, titles, & banners.

MODULE 05: Faculty Management Portal
- Coursework publishing with section targeting.
- AI-assisted assignment drafting with starter code.
- Student submission review queue with rubric grading.

MODULE 06: Campus Bulletin & Repository
- Official notice broadcasts with category tags & urgency.
- Course syllabus, lab manuals & lecture slide downloads.
- Cohort-wide student leaderboard rankings.

09 IMPLEMENTATION & RESULTS

SYSTEM IMPLEMENTATION & WORKING SCREENS

NexusEdu AI: Implementation Modules & Live Results Showcase

1. Dual Matrix Activity Heatmap (90-Day Sync)

- Unified LeetCode (⚡) & GitHub (▣) day tiles
- 7-column calendar with month navigator tabs
- Viewport floating tooltip portal

VERIFIED LIVE ✓

2. AI Code Arena (Roast & Big-O Optimizer)

- AST & loop recursion pattern detection
- O(2^N) -> O(N) automated Big-O conversion
- 5 adversarial test case generation

VERIFIED LIVE ✓

3. FAANG Mock Interview Simulator

- Company-specific rounds (Google, Meta, Amazon)
- Real-time bar-raiser logic & Big-O grading
- Instant XP & ByteCoin gamified payouts

VERIFIED LIVE ✓

4. Faculty Management & Grading Queue

- Section-targeted coursework publishing
- Integrated AI rubric evaluation
- Campus bulletin circular broadcasting

VERIFIED LIVE ✓

VERIFIED DEPLOYMENT METRICS

- **Local Server Status:**
  ACTIVE on http://127.0.0.1:5000
- **Multi-Month Calendar:**
  90 Days Active (Aug-Oct 2026)
- **AI Engine Response:**
  < 250ms Local AST / Gemini 2.0
- **Preloaded Data State:**
  Clean baseline: 0 fake records
- **Verified Accounts:**
  Faculty (ID: faculty@12345)
  Student (funwithcode8630@...)
- **GitHub Repository:**
  https://github.com/CodingWithAman1/NexusEdu-AI

10 TESTING & VALIDATION

SYSTEM TESTING & VERIFICATION MATRIX

Test ID	Test Scenario	Input Parameters	Expected vs Actual Result	Status
TC-01	Student Authentication	email: funwithcode8630@gmail.com, pass: password123	Session cookie issued; student portal opens. Actual: HTTP 200, role authenticated.	PASS ✓
TC-02	Session Bypass Guard	Access /api/assignments with unauthenticated session	Access denied with HTTP 401/403. Actual: Blocked by @login_required decorator.	PASS ✓
TC-03	Dual Calendar Multi-Month	GET /api/stats/unified-calendar (90 days sync)	90 calendar days populated across 3 months. Actual: 90 days, 0 broken HTML quotes.	PASS ✓
TC-04	AI Code Arena Big-O	POST /api/ai/code-arena with recursive fib(n)	Detects O(2^N) exponential time; rewrites to O(N). Actual: O(2^N)->O(N) generated.	PASS ✓
TC-05	FAANG Mock Interview	POST /api/ai/mock-interview (Google DP round)	Evaluates complexity & invariants; awards XP & coins. Actual: LEANING HIRE (+80 XP).	PASS ✓
TC-06	Faculty Access Separation	Student user accessing /faculty portal endpoint	Access restricted; redirect to student portal. Actual: HTTP 403 Forbidden enforced.	PASS ✓

Testing Methodology: Multi-tier unit testing with Python test harness (scratch/test_ai_suite.py), functional penetration testing, security session validation, and cross-browser execution on Chromium (Chrome/Edge).

11 CONCLUSION & ROADMAP

CONCLUSION & FUTURE SCOPE

1. PROJECT CONCLUSION

- **Successful Technical Delivery:**
Constructed an end-to-end full-stack campus platform connecting university academics with live competitive programming telemetry.
- **All Objectives Satisfied:**
Delivered 90-day dual synchronization, multimodal AI diagnostic engine, gamified ByteCoin economy, and faculty evaluation workflow.
- **Security & Documentation Hardening:**
Eliminated demo bypass vulnerabilities, reinforced strict session guards, and cleaned all dummy database entries for production baseline.
- **Key Team Learning:**
Acquired practical expertise in full-stack REST API development, SQLite normalization, session-based access control, and resilient AI fallback architectures.

2. FUTURE SCOPE & ENHANCEMENTS

- **Mobile Application (PWA / React Native):**
Develop cross-platform iOS & Android clients delivering push notifications for assignment deadlines and streak alerts.
- **Cloud Orchestration & Docker Containerization:**
Package the Flask application, Gunicorn WSGI, and SQLite database into Docker containers for seamless Kubernetes deployment.
- **Automated Plagiarism Detection (AST):**
Incorporate Abstract Syntax Tree (AST) code fingerprinting to detect unauthorized collaboration on student coursework.
- **ML-Powered Placement Probability Model:**
Train a supervised regression model predicting campus placement probability based on student GPA, LeetCode rank, and GitHub commit velocity.

12 REFERENCES

REFERENCES & BIBLIOGRAPHY (APA STYLE)

CITED ACADEMIC & TECHNICAL SOURCES:

[1] Grinberg, M. (2018). Flask Web Development: Developing Web Applications with Python (2nd ed.). O'Reilly Media.
[2] Google Cloud. (2026). Gemini API Documentation and Model Architecture. Google Developers. https://ai.google.dev
[3] MDN Web Docs. (2026). HTTP Cookies and Secure Session Management. Mozilla Corporation. https://developer.mozilla.org
[4] Cormen, T. H., Leiserson, C. E., Rivest, R. L., & Stein, C. (2022). Introduction to Algorithms (4th ed.). MIT Press.
[5] SQLite Development Team. (2026). SQLite Database Engine Architecture & Transaction Safety. https://www.sqlite.org
[6] Tailwind Labs. (2026). Tailwind CSS: Utility-First CSS Framework. https://tailwindcss.com
[7] Firebase. (2026). Firebase Authentication and Identity Management. Google. https://firebase.google.com



THANK YOU!

Any Questions or Suggestions?

PROJECT REPOSITORY & LIVE DEMO

GitHub: https://github.com/CodingWithAman1/NexusEdu-AI

Local Portal: http://localhost:5000 | Email: funwithcode8630@gmail.com

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